



Year 11 Mathematics Extension 1
Assessment Task 1
Term 2, Week 6, 2022

Name:

Teacher:

Monday 30th May, 2022

Setter: WIL

General Instructions

- **Working time – 45 minutes.**
- Answer all questions on the lined paper provided using black or blue pen.
- NESA approved calculators may be used.
- Show all relevant mathematical reasoning and/or calculations.
- Please remove the last page and submit it with your solutions for Questions 3.
- **Total marks – 32.**
- Attempt Questions 1 – 6.

Question 1 (5 marks)		Marks
(a)	(i) Show that $(x-3)$ is a factor of the polynomial $P(x) = x^3 - 4x^2 + x + 6$.	1
	(ii) Hence or otherwise, express $P(x)$ in fully factored form.	1
(b)	Consider the functions $f(x) = \frac{1}{x+1}$ and $g(x) = \frac{1}{1-x}$.	
	(i) Show that $f \circ g(x) = \frac{1-x}{2-x}$.	1
	(ii) Without the use of calculus, sketch (using quarter of a page) $y = f \circ g(x)$ showing all the intercepts with the axes, asymptotes and any exclusions in its natural domain.	2

End of Question 1

Question 2 (5 marks) (Start a new page)

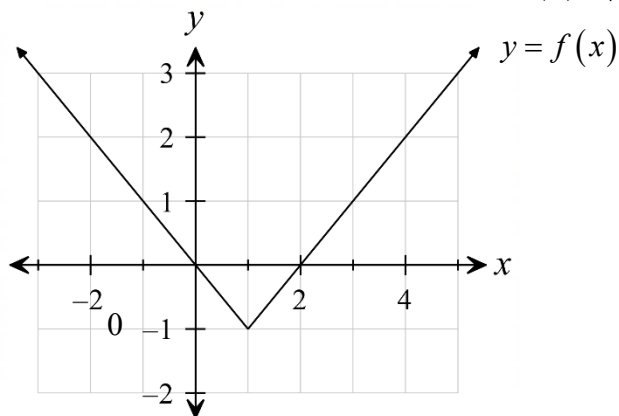
Marks

- (a) Write the Cartesian equation given by the parametric equations $x = 4 + \operatorname{cosec} \theta$ and $y = 2 + \cot \theta$. **2**
- (b) The roots of $x^3 - x^2 - 5x + 2 = 0$ are α, β and γ .
- (i) Show that $\beta + \gamma = 1 - \alpha$. **1**
- (ii) Using part (i), or otherwise, evaluate $\frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta} + \frac{\alpha + \beta}{\gamma}$. **2**

End of Question 2

Remove pages 8 and 9 (last page) from your test paper and use them to sketch the graphs and answer Question 3. When sketching graphs, show all intercepts with axes, asymptotes and changes to the curves' shape. Please go over $f(x)$ on the template with your pen to attract marks where necessary.

- (a) The diagram below shows the graph of $f(x) = |x-1| - 1$.

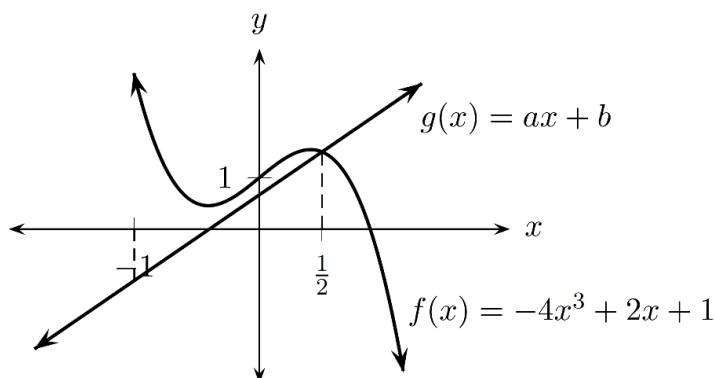


Sketch each of the following using the templates supplied on page 7:

(i) $y = \sqrt{f(x)}$ 1

(ii) $y = \frac{1}{f(x)}$ 1

- (b) The graphs of $f(x) = -4x^3 + 2x + 1$ and $g(x) = ax + b$ are shown below.



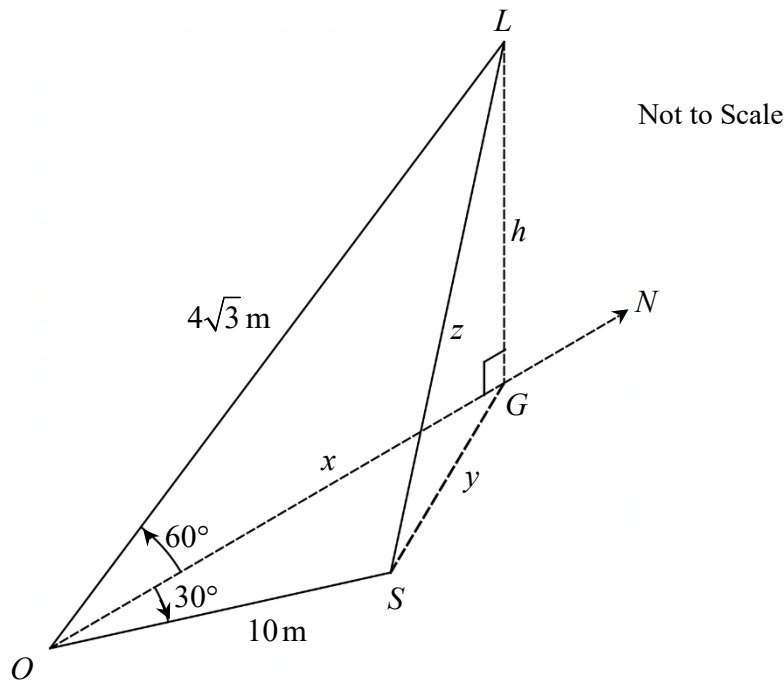
- (i) Sketch the graph for $h(x) = -4|x|^3 + 2|x| + 1$ using the template supplied on page 8. 1

- (ii) The solution to the inequality $ax + b > -4|x|^3 + 2|x| + 1$ is $x \in (-\infty, -1) \cup \left(\frac{1}{2}, \infty\right)$.

Find the values of a and b . 2

End of Question 3

- (a) During the Festival of Lanterns, Owen releases a lantern from point O . It floats up into the sky due north in a straight line with an angle of elevation of 60° . Sam is located at S , 10 metres from Owen on a bearing of $N30^\circ E$. The lantern has travelled $4\sqrt{3}$ metres from O when Sam first observes it at L . The point G is on the ground directly below L .



Given that $OG = x$, $GL = h$, $SG = y$ and $SL = z$. Find the distance SL , the distance between Sam and the lantern. Leave your answer correct to two decimal places.

3

- (b) The function $g(x) = x^2 - 6x + 8$ has been restricted to the domain $x \leq 3$, so that it has an inverse function. By the use of the quadratic formula, or otherwise, find the equation of the inverse function $g^{-1}(x)$.

2

End of Question 4

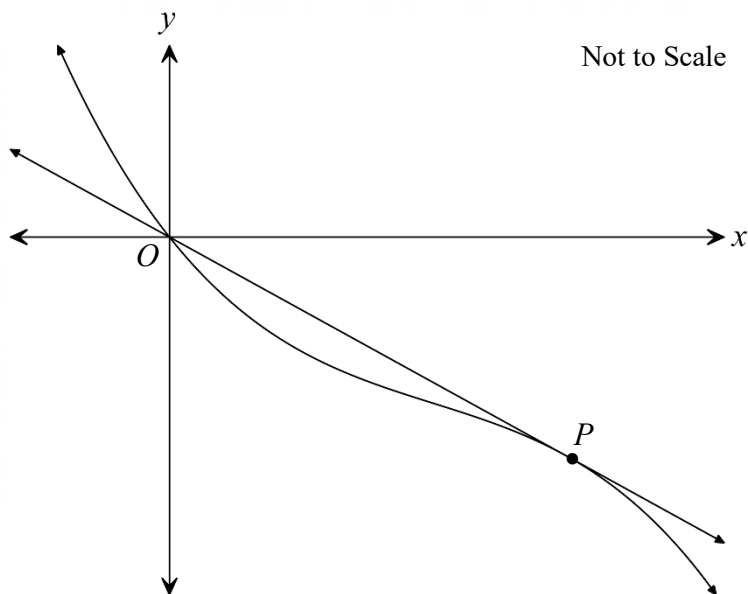
Question 5 (6 marks) (Start a new page)

Marks

- | | | | |
|-----|-------|---|----------|
| (a) | (i) | Solve the inequality $\frac{x}{1-x^2} \geq 0$ for x . | 3 |
| | (ii) | Show that $\sec x \tan x = \frac{\sin x}{1-\sin^2 x}$. | 1 |
| | (iii) | Hence, or otherwise, find all solutions of $\sec x \tan x \geq 0$ for $0 \leq x \leq 360^\circ$. | 2 |

End of Question 5

- (a) The line $y = mx$ is a tangent to the curve $y = -x^3 + 3x^2 - 4x$ at the point P .



Determine the the coordinates of the point P and the equation of the tangent.

3

- (b) The polynomial $P(x) = px^3 + qx + s$ has a double root at $x = -1$ and a remainder is 6 when it is divided by $(x-1)$. If p , q and s are real, find their values.

3

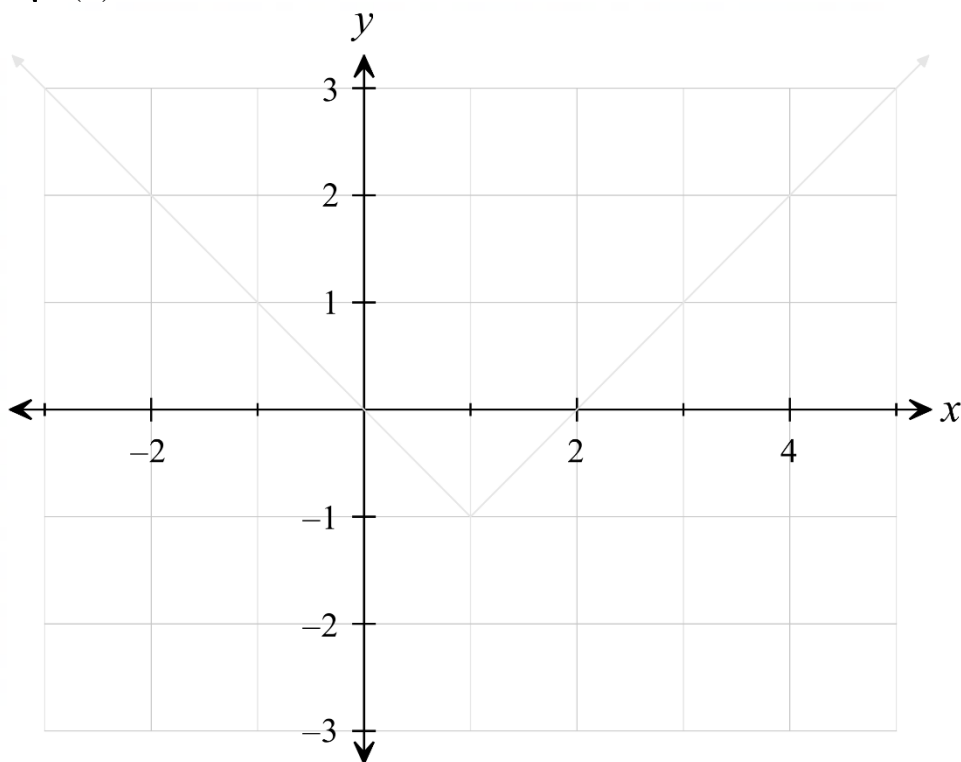
End of Assessment Task

Name:

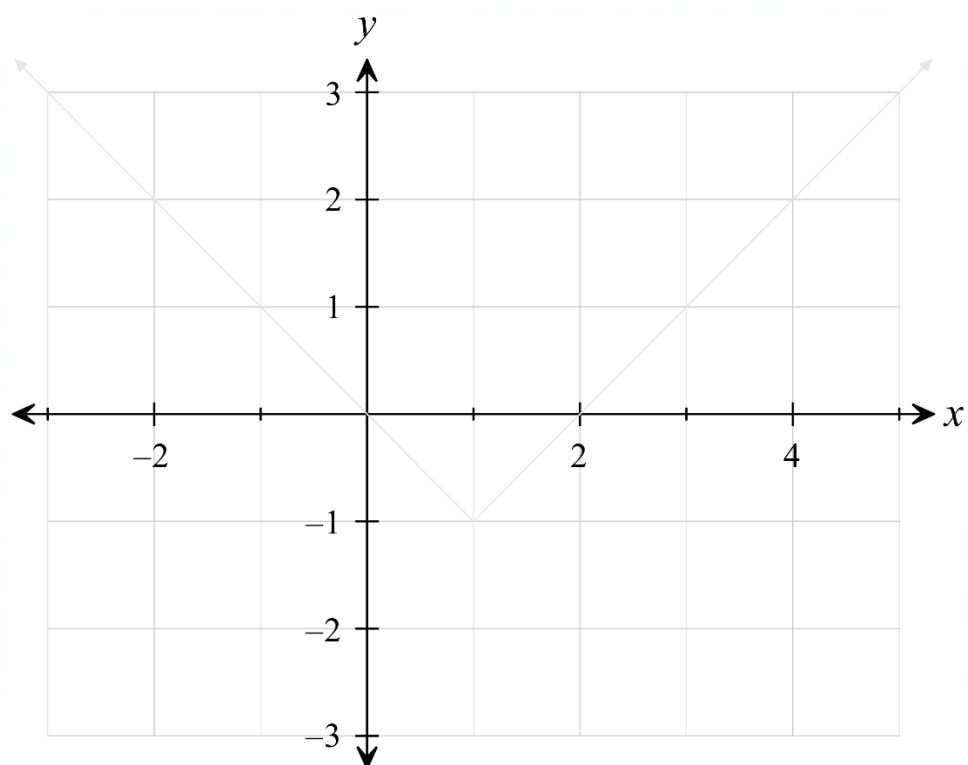
Teacher:

Answer Sheet for Question 3 (a) (Use the light grey as a guide, only lines made in pen will be marked).

(a) (i) $y = \sqrt{f(x)}$



(ii) $y = \frac{1}{f(x)}$

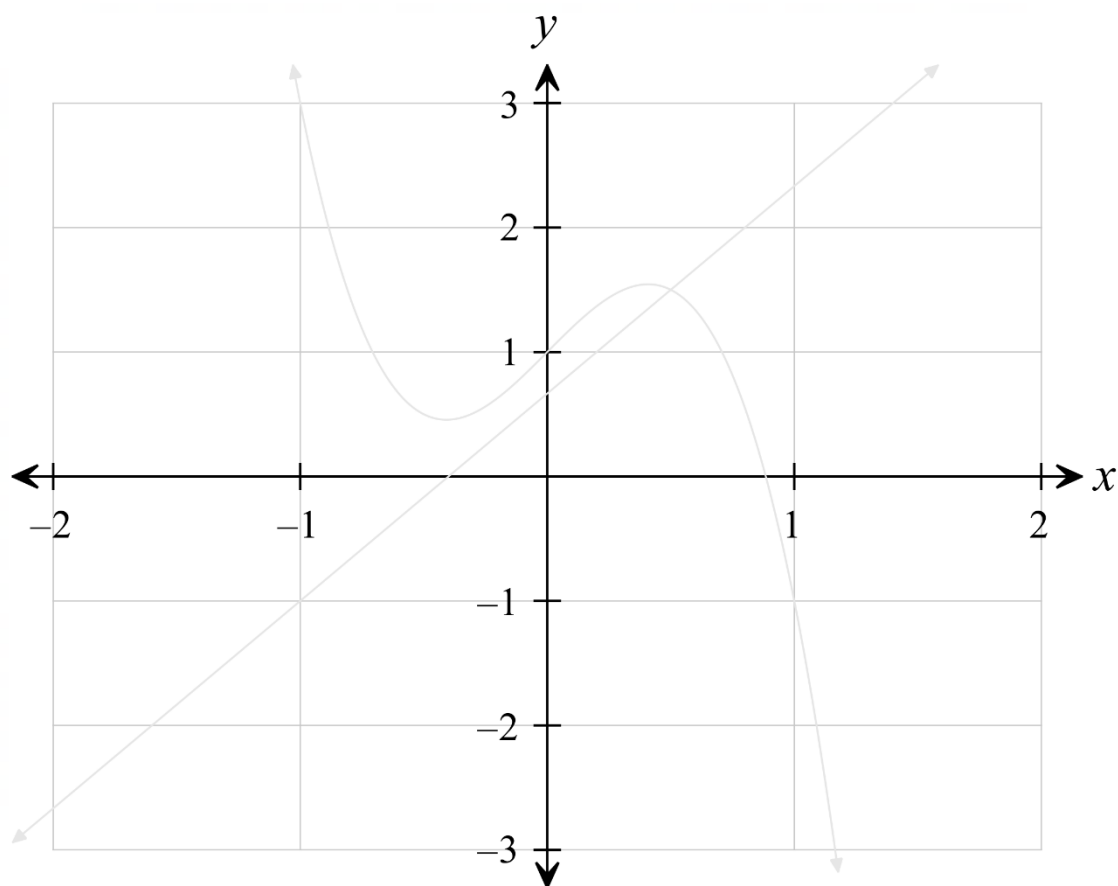


Name:

Teacher:

Answer Sheet for Question 3 (b) (Use the light grey as a guide, only lines made in pen will be marked).

(b) (i) $h(x) = -4|x|^3 + 2|x| + 1$



(ii) Write your answer here.



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Suggested Solution (s)	Comments
<u>Question 1.</u> (a)(i) $P(3) = 3^3 - 4(3^2) + 3 + 6$ $= 27 - 36 + 3 + 6$ $= 0$ (ii) <u>method 1</u> $\begin{array}{r} x^2 - x - 2 \\ x-3 \overline{) x^3 - 4x^2 + x + 6} \\ \underline{-(x^3 - 3x^2)} \\ -x^2 + x \\ \underline{-(-x^2 + 3x)} \\ -2x + 6 \\ \underline{-(-2x + 6)} \\ 0 \end{array}$ and then factor quotient $(x-2)(x+1)$ <u>method 2</u> look at possible factors considering the constant term. try $P(2)$ and then $P(-1)$ or $P(-2)$ and $P(1)$ Final answer: $P(x) = (x-3)(x-2)(x+1)$ ✓ (b)(i) $f \circ g(x) = \frac{1}{\frac{1}{1-x} + 1}$ $= \frac{1}{\frac{1 + 1 - x}{1-x}}$ $= \frac{1-x}{2-x}$ $= \frac{2-x-1}{2-x}$ $= 1 - \frac{1}{2-x}$ $= 1 + \frac{1}{x-2}$ at $x=1$ $f \circ g(x) = 0$ at $x=0$ $f \circ g(x) = \frac{1}{2}$ (ii)	either line for one mark. (i) one mark for correct substitution. (ii) key inclusions • y-intercept • exclusions at $x=1$ • shape in correct quadrants • x and y asymptotes all 4 required for 2 marks 2 required for one mark.

Question 2.

$$(a) \left. \begin{aligned} (x-4)^2 &= \operatorname{cosec}^2 \theta \\ (y-2)^2 &= \cot^2 \theta \end{aligned} \right\} \text{ both 1 mark.}$$

...recognising $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$ no mark alone.

by subtraction:

$$(x-4)^2 - (y-2)^2 = \operatorname{cosec}^2 \theta - \cot^2 \theta$$

$$(x-4)^2 - (y-2)^2 = 1 \quad \checkmark$$

(b) (i) investigating the sum gives

$$\alpha + \beta + \gamma = 1 \quad \checkmark$$

$$\therefore \beta + \gamma = 1 - \alpha \text{ as required}$$

(ii) rearranging this expression also gives

$$\alpha + \beta = 1 - \gamma \text{ and } \alpha + \gamma = 1 - \beta$$

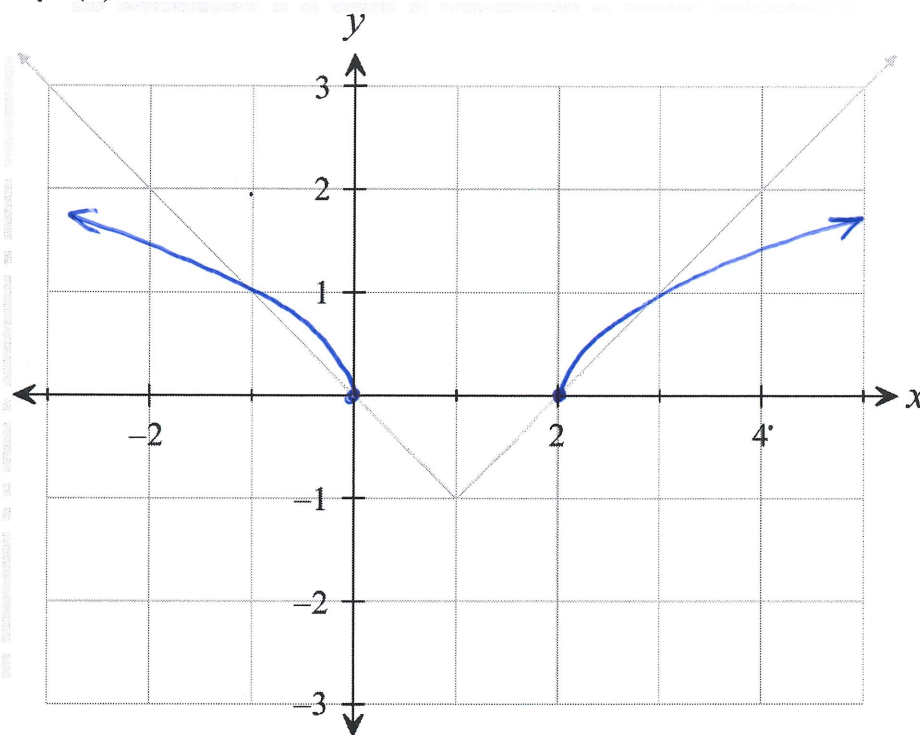
$$\begin{aligned} \text{so that: } & \frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta} + \frac{\alpha + \beta}{\gamma} \\ &= \frac{1 - \alpha}{\alpha} + \frac{1 - \beta}{\beta} + \frac{1 - \gamma}{\gamma} \\ &= \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} - 3 \quad \checkmark \\ &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} - 3 \\ &= \frac{-5}{-2} - 3 \\ &= -\frac{1}{2} \quad \checkmark \end{aligned}$$

Name: Solutions

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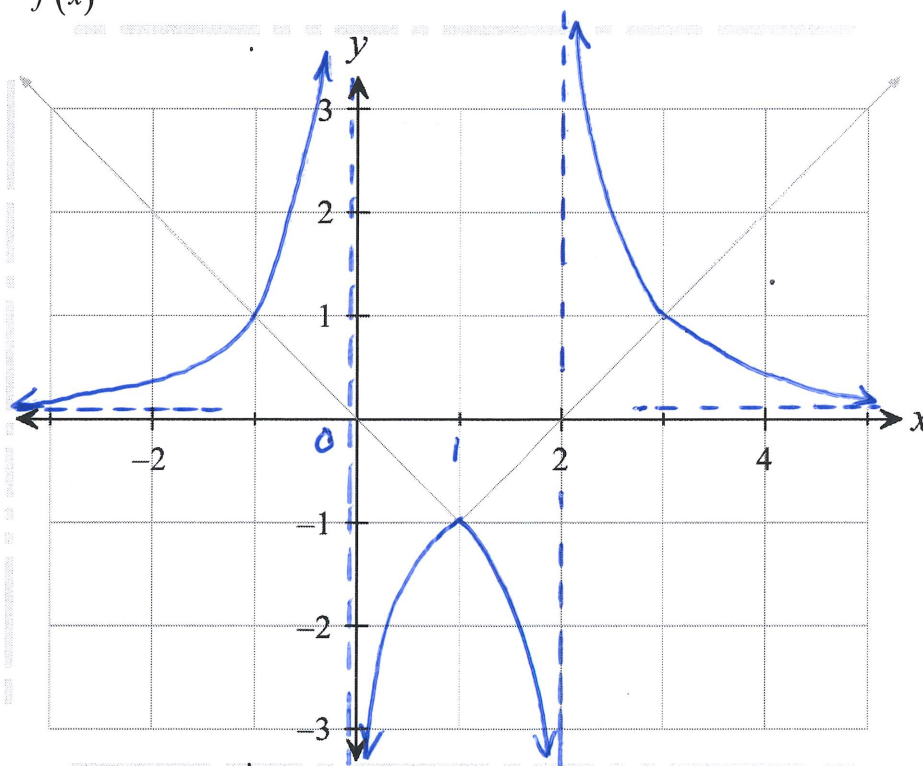
Answer Sheet for Question 3 (a) (Use the light grey as a guide, only lines made in pen will be marked).

(a) (i) $y = \sqrt{f(x)}$



- lines must cut $(-1, 1)$ and $(3, 1)$
- must be above $f(x)$ for $(-1, 0)$ and $(2, 3)$ and below for $(-\infty, -1)$ and $(3, \infty)$

(ii) $y = \frac{1}{f(x)}$



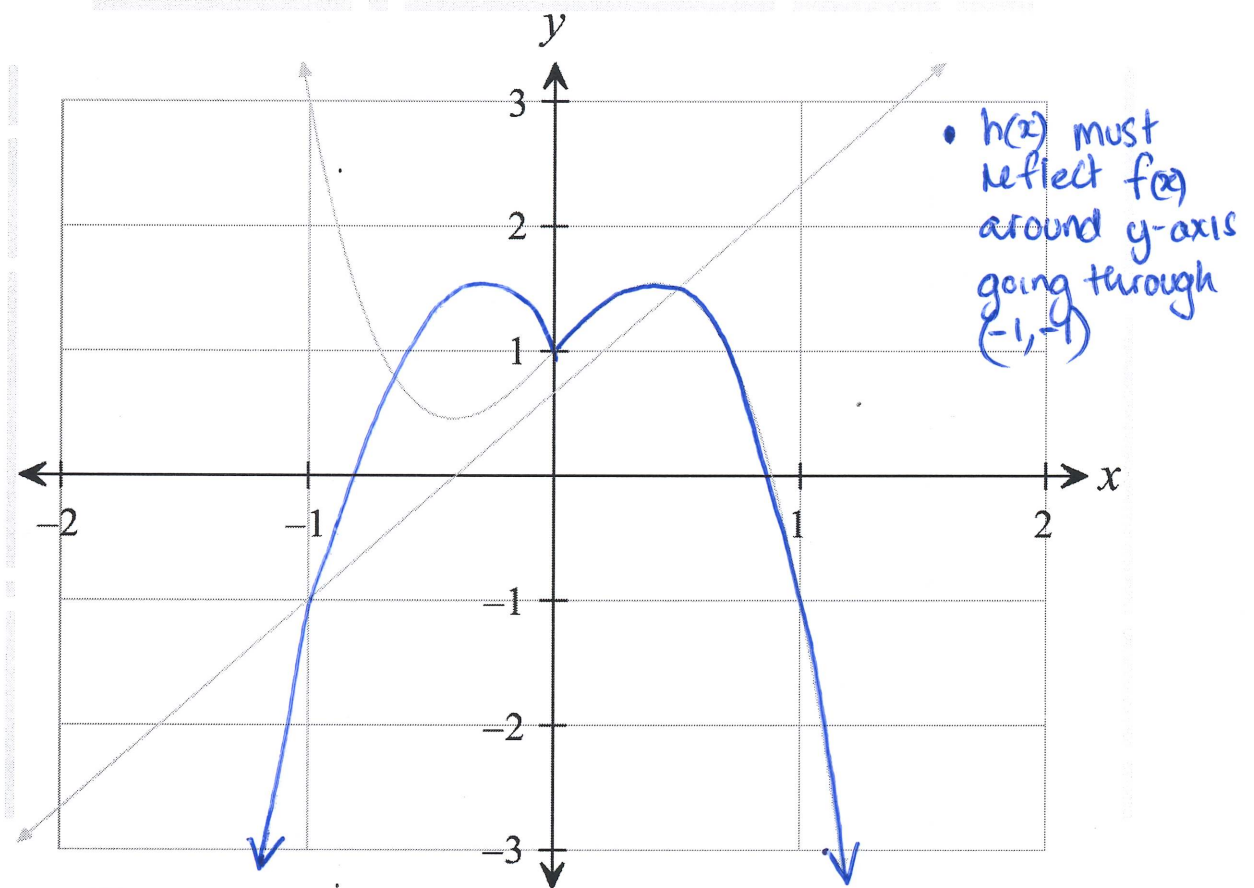
- lines must cut at $(-1, 1)$ and $(3, 1)$
- show vert. asymptotes of $x=0, x=2$
- horizontal asymptote at $y=0$
- nature of graph at $x=1$

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Answer Sheet for Question 3 (b) (Use the light grey as a guide, only lines made in pen will be marked).

(b) (i) $h(x) = -4|x|^3 + 2|x| + 1$



(ii) P and Q are the pts of intersection of $g(x)$ and $h(x)$

$$h\left(\frac{1}{2}\right) = -4\left(\frac{1}{2}\right)^3 + 2\left|\frac{1}{2}\right| + 1 = \frac{3}{2} \quad \therefore P\left(\frac{1}{2}, \frac{3}{2}\right)$$

$$h(-1) = -4|-1|^3 + 2|-1| + 1 = -1 \quad \therefore Q(-1, -1)$$

using gradient formula $a = \frac{-1 - \frac{3}{2}}{-1 - \frac{1}{2}} = \frac{5}{3} \checkmark$

substituting $Q(-1, -1)$ into $g(x)$ to find b

$$-1 = \frac{5}{3}(-1) + b$$

$$b = \frac{5}{3} - 1$$

$$b = \frac{2}{3} \checkmark$$



	Comments
<u>Question 4.</u> (a) $SL = Z$, we know that $z^2 = y^2 + h^2$ from ΔSGL and $y^2 = x^2 + 10^2 - 2 \times 10 \times x \cos 30^\circ$ $\therefore z^2 = x^2 + 10^2 - 10\sqrt{3}x + h^2$ $\sin 60 = \frac{h}{4\sqrt{3}} \therefore h = 6$ $\cos 60 = \frac{x}{4\sqrt{3}} \therefore x = 2\sqrt{3}$ $z^2 = (2\sqrt{3})^2 + 10^2 - 10\sqrt{3} \times 2\sqrt{3} + 6^2$ $= 12 + 100 - 60 + 36$ $= 88 \therefore z = \sqrt{88} \text{ or } 2\sqrt{22}$ $z = 9.38 \text{ m} \checkmark$ (b) inverse relation is $x = y^2 - 6y + 8$ <u>method 1</u> $y^2 - 6y + 8 - x = 0$ <u>method 2</u> (completing the square). $y = \frac{6 \pm \sqrt{36 - 4(1)(8-x)}}{2}$ $x - 8 + 9 = y^2 - 6y + 9$ $y = \frac{6 \pm \sqrt{4(1+x)}}{2}$ $x + 1 = (y - 3)^2$ $y = 3 \pm \sqrt{1+x}$ $y - 3 = \pm \sqrt{1+x}$ $y = 3 \pm \sqrt{1+x}$ $y = 3 \pm \sqrt{1+x}$ but domain of $g(x)$ is $x \leq 3$ $\therefore$ range of $g^{-1}(x)$ is $y \leq 3$ $\therefore g^{-1}(x) = 3 - \sqrt{1+x} \checkmark$	correctly recognising and utilising Pythag and cosine rule $\Rightarrow$ 1 mark. correct values of both x and $h \Rightarrow$ one mark. either method one mark. students must consider the range/domain for second mark.



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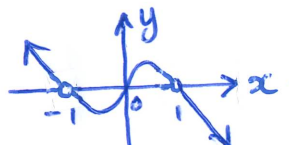
Comments

Question 5.

(a) (i) $\frac{x}{1-x^2} \times (1-x^2)^2 \geq (1-x^2)^2 \times 0 \quad x \neq \pm 1$

$$x(1-x^2) \geq 0$$

$$x(1-x)(1+x) \geq 0$$

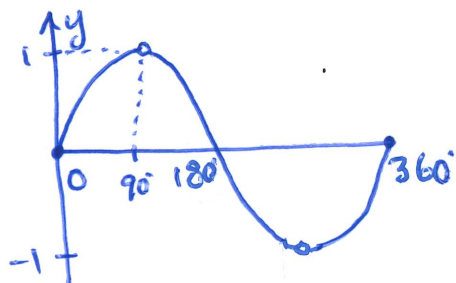


$$\left. \begin{array}{l} x < -1 \text{ or } 0 \leq x < 1 \\ (-\infty, -1) \cup [0, 1) \end{array} \right\} \text{either}$$

(ii) $\sec x \tan x = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$
 $= \frac{\sin x}{\cos^2 x} \checkmark$
 $= \frac{\sin x}{1 - \sin^2 x} \quad \because \sin^2 x + \cos^2 x = 1$

(iii) $\frac{\sin x}{1 - \sin^2 x} \geq 0$ will give same solution as $\sec x \tan x \geq 0$ (part (ii))

from part (i) $\sin x < -1$ or $0 \leq \sin x < 1$
 $\downarrow$
no solution



$$0 \leq x < 90^\circ \text{ or } 90^\circ < x \leq 180^\circ \text{ or } x = 360^\circ$$

one mark for correct method.

2 marks for considering exclusion of indomain

2 marks all 3 solutions and 1 mark for 2 correct solutions



2022 Year 11 AT1 Mathematics Extension 1 - SOLUTIONS

Suggested Solution (s)	Comments
<u>Question 6.</u> (a) $mx = -x^3 + 3x^2 - 4x$ rearranging... $-x^3 + 3x^2 - (4+m)x = 0$ factoring... $x(x^2 - 3x + (4+m)) = 0$ ✓ $x = 0$ correlates to passing through origin. so, using sum and product of roots: let roots be α, α (only one root $\Rightarrow$ tangent) sum: $2\alpha = 3$ $\alpha = \frac{3}{2}$ $y = -\left(\frac{3}{2}\right)^3 + 3\left(\frac{3}{2}\right)^2 - 4\left(\frac{3}{2}\right)$ $y = -\frac{27}{8} + \frac{27}{4} - \frac{12}{2}$ $y = -\frac{21}{8} \therefore P\left(\frac{3}{2}, -\frac{21}{8}\right)$ or using $\Delta = 0$, $(-3)^2 - 4(4+m) = 0$ $9 - 16 - 4m = 0$ $-7 - 4m = 0$ $m = -\frac{7}{4}$ So $x^2 - 3x + \left(4 - \frac{7}{4}\right) = 0$ $4x^2 - 12x + 9 = 0 \therefore (2x-3)^2 = 0 \Rightarrow x = \frac{3}{2}$ $P\left(\frac{3}{2}, -\frac{21}{8}\right)$ if using method ① $\left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right) + 4 + m = 0$ $\frac{9}{4} - \frac{9}{2} + 4 + m = 0$ $m = -\frac{7}{4}$ $\therefore$ tangent $y = -\frac{7}{4}x$ ✓ (b) $P(-1) = P'(-1) = 0$ and $P(1) = 6$ $P(-1) = -p - q + s$ $P'(x) = 3px^2 + q$ $P'(-1) = 3p + q$ $\therefore 3p + q = 0$ — ① $P(-1) = -p - q + s = 0$ $p + q - s = 0$ — ② $P(1) = p + q + s = 6$ — ③ solve ①, ② and ③ simultaneously	one mark for this factored form. one for either method. 1 mark for forming 2 eqns correctly. 2 marks for all correct and making attempt to solve correctly.

$$\textcircled{2} + \textcircled{3}: p + q = 3 \text{ — } \textcircled{4}$$

$$\textcircled{1} - \textcircled{4} \quad 2p = -3$$

$$p = -\frac{3}{2}$$

$$\text{substitute } p \text{ into } \textcircled{4}$$

$$q = 3 + \frac{3}{2}, q = \frac{9}{2}$$

$$s = \frac{9}{2} - \frac{3}{2}, s = 3$$

3 marks for finding p, q and s correctly.